

Geometric versus classical integrators for Schwarzschild geodesics: error growth, invariants and cost over short and long integrations

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Numerical geodesics underlie black-hole imaging and the modelling of extreme-mass-ratio inspirals, yet the choice of integrator is often made by habit. We compare eight integrators on timelike and null geodesics of the Schwarzschild metric: the classical Runge–Kutta method (RK4), the embedded pairs DP5(4) and DOP853, the Gauss–Legendre collocation methods of order 2, 4 and 6, and Tao’s explicit symplectic scheme of order 2 and 4. Each is applied to the second-order geodesic equation and to Hamilton’s equations, and every error is measured against a closed-form reference built from elliptic integrals and functions. Over 10^4 radial periods the Runge–Kutta methods drift linearly in the mass shell and quadratically in phase, while the symplectic methods keep the mass-shell error bounded and the phase error linear; sixth-order Gauss–Legendre reaches round-off and then follows Brouwer’s $N^{1/2}$ law. Gauss–Legendre applied to the second-order system, where it is symmetric but not symplectic, conserves as well as in canonical variables. For short integrations DOP853 is the cheapest method at every accuracy; over 1000 periods only sixth-order Gauss–Legendre reaches a phase error of 10^{-8} . On the photon sphere every method survives at most about six orbits, gaining $\ln 10/2\pi$ orbits per decade of accuracy as predicted, and plain floating-point summation fakes a longer survival. At the ISCO the time to plunge grows algebraically, as $\varepsilon^{-1/4}$ in the size of the integration error, and the sign of the error decides whether the orbit plunges at all. Every figure and table is regenerated from a clean checkout by one command.

I. INTRODUCTION

Geodesics of black-hole spacetimes are integrated numerically in two very different regimes. Ray tracing for images of accretion flows, such as those behind the Event Horizon Telescope results [1], follows millions of photons for a short affine time each; codes such as GRay [2], geokerr [3], RAPTOR [4] and GYOTO [5] are built around adaptive Runge–Kutta methods or semi-analytic solutions. The evolution of bound orbits, as in the adiabatic modelling of extreme-mass-ratio inspirals, follows a few orbits for 10^3 – 10^5 revolutions, where the long-time behaviour of the integration error matters more than its size after one step. Geometric integrators [6] are designed for the

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second regime, and several have been adapted to geodesics: Gauss collocation with a symmetric step-size control [7], explicit symplectic splittings for Schwarzschild [8], and Tao’s extended phase space [9, 10], which the FANTASY code applies to arbitrary metrics [11].

Comparisons between these families are usually made on one test problem, one formulation and one length of integration. This report compares them systematically, on the simplest spacetime where every relevant quantity has a closed form. We ask five research questions:

RQ1: How do the mass-shell error and the phase error grow over long integrations?

RQ2: Does the formulation, the second-order geodesic equation or Hamilton’s equations, change which quantities are conserved, and is a symmetric method that is not symplectic in the chosen variables enough?

RQ3: Which method reaches a given accuracy at the lowest cost, and how does the answer depend on the length of the integration?

RQ4: How far can an integrator extend the life of an orbit on the unstable photon sphere, and of an orbit at the marginally stable ISCO?

RQ5: Which integrator should one use for ray tracing, and which for long orbit evolution?

The study has four design choices that the results depend on. First, both formulations keep the full eight-dimensional phase space, so that the conservation of the energy E and the axial angular momentum L_z can be tested rather than imposed. Second, the integrators see only a vector field, so the same code runs on toy problems, on Schwarzschild and, later, on Kerr. Third, the cost of a run is the number of vector-field evaluations, counting every fixed-point iteration of the implicit methods. Fourth, errors are measured on physical quantities against exact references: the deflection angle at the radius where the integration stops, the angle swept per radial period and the orbit $r(\phi)$ through elliptic functions, and the exit time from the photon sphere through an elliptic integral.

Section II sets out the two formulations and their invariants, Sec. III the integrators and Sec. IV the test problems and references. Section V presents the results, ordered by research question. Section VI turns them into recommendations, and Sec. VII concludes.

II. TWO FORMULATIONS OF THE GEODESIC PROBLEM

We use the signature $(-, +, +, +)$, units $G = c = M = 1$ and Schwarzschild coordinates $x^\mu = (t, r, \theta, \phi)$ with $f(r) = 1 - 2/r$. A dot is the derivative with respect to the affine parameter λ , and $\epsilon = 1$ for timelike geodesics ($\lambda = \tau$), $\epsilon = 0$ for null ones (normalized by $E = 1$, so that $L = b$).

a. (a) *Second order.* With $u^\mu = \dot{x}^\mu$ the geodesic equation becomes

$$\ddot{x}^\mu = u^\mu, \quad \dot{u}^\mu = -\Gamma_{\alpha\beta}^\mu(x) u^\alpha u^\beta, \quad g_{\mu\nu} u^\mu u^\nu = -\epsilon, \quad (1)$$

a system for $y = (x^\mu, u^\mu) \in \mathbb{R}^8$. The nine independent Christoffel symbols are listed in Appendix A.

b. (b) *Hamiltonian.* The Legendre transform of the Lagrangian $\mathcal{L} = \frac{1}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$, with momenta $p_\mu = g_{\mu\nu} \dot{x}^\nu$, gives the super-Hamiltonian [12]

$$H = \frac{1}{2} g^{\mu\nu}(x) p_\mu p_\nu = \frac{1}{2} \left[-\frac{p_t^2}{f} + f p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\phi^2}{r^2 \sin^2 \theta} \right], \quad H = -\frac{\epsilon}{2}, \quad (2)$$

with $\dot{x}^\mu = g^{\mu\nu} p_\nu$ and $\dot{p}_\mu = -\frac{1}{2} \partial_\mu g^{\alpha\beta} p_\alpha p_\beta$, for $y = (x^\mu, p_\mu)$. Only $g^{\mu\nu}$ and its first derivatives enter, so (b) needs no Christoffel symbols. H is not separable: the kinetic term depends on position, which rules out the explicit leapfrog family.

c. *Invariants.* Both systems conserve the energy, the axial and total angular momentum and the mass shell:

$$E = f u^t = -p_t, \quad L_z = r^2 \sin^2 \theta u^\phi = p_\phi, \quad L^2 = r^4 \left[(u^\theta)^2 + \sin^2 \theta (u^\phi)^2 \right] = p_\theta^2 + \frac{p_\phi^2}{\sin^2 \theta}. \quad (3)$$

In (b), t and ϕ are cyclic, so $\dot{p}_t$ and $\dot{p}_\phi$ are the floating-point number zero. Every Runge–Kutta method, explicit or implicit, then conserves E and L_z bit for bit, whatever its order, step size or number of iterations. This is a property of the variables, not a merit of the method. Informative tests of E and L_z are therefore made in (a), where they are nonlinear functions of the state; in (b) the informative invariants are H and, for inclined orbits, L^2 , the Schwarzschild limit of Carter’s constant ($Q = L^2 - L_z^2$). None of the invariants is quadratic in the state, so the exact conservation of quadratic invariants by Gauss methods does not apply; symplectic methods can only keep their errors bounded [6]. We report relative errors $\delta H = |H + \frac{1}{2}|/\frac{1}{2}$ (timelike), $|H|/E^2$ (null), and $\delta E = |E - E_0|/E_0$, and likewise for L_z and L^2 .

d. *Symplecticity and reversibility.* Formulation (b) is canonical; a symplectic method applied to it produces symplectic maps, and backward error analysis bounds the energy error over exponentially long times [6]. Formulation (a) describes the same flow in the non-canonical variables (x, u) , where the symplectic form has x -dependent coefficients; a Gauss method applied to (a) is

TABLE I. The integrators (T1). For the implicit methods the cost per step is the number of stages times the number of fixed-point iterations; the measured values are averages over 10^4 periods of the orbit $(p, e) = (20, 0.5)$ at 1000 steps per period.

Method	Structure	Order	Sympl.	Symm.	Adapt.	Evaluations per step	Form.
RK4	explicit Runge-Kutta	4	no	no	no	4	(a), (b)
DP5	explicit embedded pair 5(4)	5	no	no	yes	6	(a), (b)
DOP853	explicit embedded pair 8(5,3)	8	no	no	yes	12	(a), (b)
GL1	implicit midpoint (Gauss, $s = 1$)	2	yes	yes	no	1 per iteration; measured 6.8 (b), 7.9 (a)	(a), (b)
GL2	Gauss collocation, $s = 2$	4	yes	yes	no	2 per iteration; measured 11.2 (b), 13.0 (a)	(a), (b)
GL3	Gauss collocation, $s = 3$	6	yes	yes	no	3 per iteration; measured 14.2 (b), 18.4 (a)	(a), (b)
Tao2	extended phase space, Strang	2	yes	yes	no	3	(b) only
Tao4	extended phase space, triple jump	4	yes	yes	no	9	(b) only

symmetric but not symplectic. Both systems are reversible with respect to $\rho(x, v) = (x, -v)$, and for integrable reversible systems symmetric methods share the long-time behaviour of symplectic ones [6, Ch. XI]. Comparing Gauss–Legendre in (a) and (b) tests whether reversibility alone suffices here (RQ2).

e. Initial data. The constraint is a quadratic in one momentum component, solved in a cancellation-free form with the sign of the velocity. Near a radial turning point $p_r = \pm\sqrt{E^2 - V_{\text{eff}}}/f$ is computed from a difference of nearly equal numbers and comes out of order $\sqrt{\epsilon_{\text{mach}}}$ instead of zero. Bound and circular orbits therefore start exactly at a turning point, with $p_r = 0$ and the constraint solved for p_t . The initial relative constraint is at most 3.3×10^{-16} on every orbit of the study, in both formulations.

III. INTEGRATORS

All integrators share one interface: they receive the vector field $F(y)$ (and, for Tao’s method, the canonical structure of the state) and know nothing about the metric. The properties of the eight methods are collected in Table I.

a. Explicit Runge–Kutta. RK4 is the classical four-stage method with a fixed step; the parameter after n steps is $\lambda_0 + nh$, never a running sum. DP5(4) [13] and DOP853 [14] are the embedded pairs of SciPy’s RK45 and DOP853, reimplemented in compiled code with SciPy’s step-size controller copied line by line. From a generic starting point both take the same steps as SciPy and use the same number of evaluations at tolerances 10^{-4} – 10^{-12} ; step sizes agree to 10^{-7} . Events (periapsis passages, crossings of a radius) are located by root finding on partial steps of the method itself,

so their accuracy matches the order of the method; linear interpolation would cap the measured order at two.

b. Gauss–Legendre. The s -stage Gauss methods GL1 (implicit midpoint), GL2 and GL3 have order $2s$, and are symplectic and symmetric [6]. The stage equations are solved by fixed-point iteration on the increments $Z_i = Y_i - y_n$, started from the collocation polynomial of the previous step. The iteration runs until the norm of the correction stops decreasing, the rule recommended for round-off-limited integrations [15], with a guard: stagnation counts only once the correction is within 100 ulp of the increments, because on rotation-like problems the correction can wobble long before round-off. The update $y_{n+1} = y_n + d^\top Z$, $d = b^\top A^{-1}$, is added with compensated summation [16]. An optional absolute tolerance on the correction replaces the stagnation rule for the experiment of Sec. V G.

c. Tao’s method. Tao [9] integrates two copies (q, p) and (x, y) of the state with

$$\tilde{H}(q, p, x, y) = H(q, y) + H(x, p) + \frac{\omega}{2} (|q - x|^2 + |p - y|^2). \quad (4)$$

The flows of the first two terms are explicit shifts, the third rotates $(q - x, p - y)$ by the angle $2\omega\delta$. The second-order method is the Strang composition $\phi_A^{h/2} \phi_B^{h/2} \phi_C^h \phi_B^{h/2} \phi_A^{h/2}$, the fourth-order one its triple jump with $\gamma_1 = 1/(2 - 2^{1/3})$, $\gamma_2 = 1 - 2\gamma_1$ [17]. Each flow is one call of the vector field of (b); caching the gradient between consecutive flows gives 3 and 9 evaluations per step. Two choices are not fixed by the method. We couple only the non-cyclic coordinates (r, θ) : coupling t and ϕ as well lets the time copies drift apart by up to 10^4 , and the rotation then feeds that difference into p_t , so E is no longer conserved. And ω must be chosen; Sec. V G fixes $\omega = 10^{-3}$ for all experiments.

d. Cost. The cost of a run is its number of vector-field evaluations n_{fev} , including every fixed-point iteration and, for Tao, every gradient. Wall times are compared only between our own compiled implementations, with one thread per process, after compilation, as the median of repeats; they track n_{fev} up to a factor of order one. One evaluation costs 0.7–1.6 μs inside the drivers on the machine of Appendix C.

e. Verification. Before any geodesic run the integrators were tested on the harmonic oscillator, the Kepler problem and Tao’s nonseparable example $H = \frac{1}{2}(q^2 + 1)(p^2 + 1)$. The measured orders are 2.00, 4.00, 6.00 for GL1–GL3, 2.00 and 3.98 for Tao2 and Tao4, and 4.05 for RK4. The one-step maps of GL and Tao preserve the symplectic form to 2×10^{-12} (the accuracy of the finite-difference Jacobian; Tao in its extended space), against 4×10^{-3} for RK4, and they are symmetric, $\Psi_{-h} \circ \Psi_h = \text{id}$, to 2×10^{-16} . Over 10^6 steps of Tao’s example their energy error is bounded, while RK4’s doubles from the first to the second half of the run.

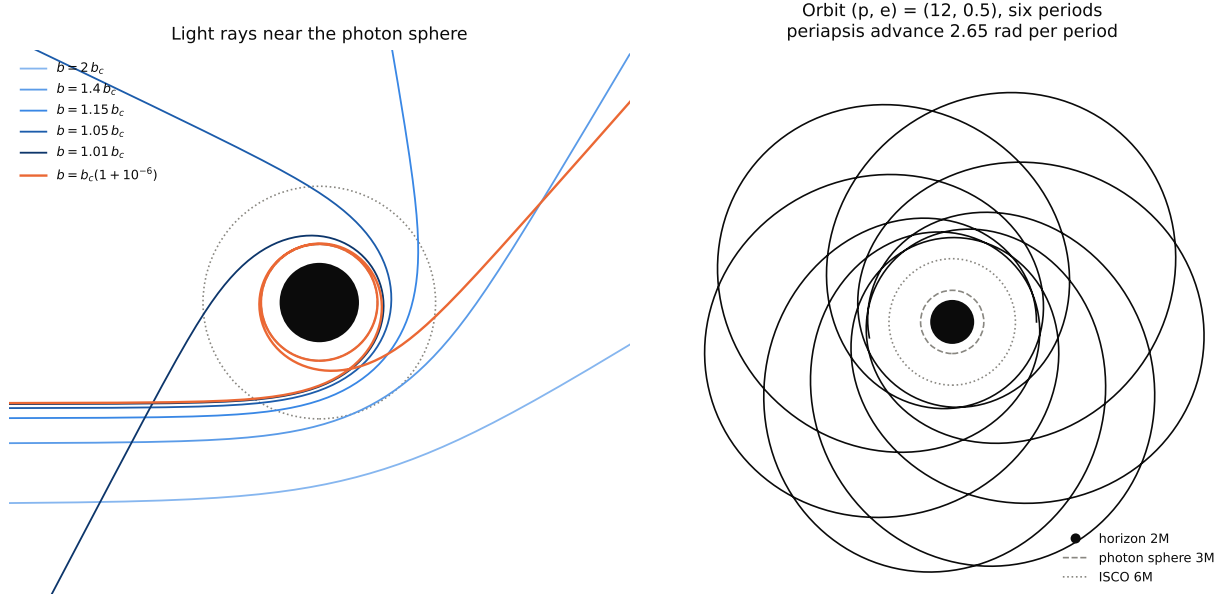


FIG. 1. The geodesics of the study in the equatorial plane, with the horizon ($r = 2$), the photon sphere ($r = 3$) and the ISCO ($r = 6$). Left: a parallel bundle of light rays with impact parameters from $2b_c$ to $b_c(1 + 10^{-6})$; the last winds about twice around the photon sphere. Right: six radial periods of the bound orbit $(p, e) = (12, 0.5)$.

IV. TEST PROBLEMS AND EXACT REFERENCES

Fig. 1 shows the geodesics of the study. Each test case fixes the initial data, the span of the integration, the events that stop it, an exact reference and an error defined on a physical quantity, never on the raw state, because the states of (a) and (b) differ.

a. TC1, light deflection. A photon with impact parameter $b > b_c = 3\sqrt{3}$ starts at r_{far} and is followed to r_{far} again. With $u = 1/r$ and the roots $u_1 < 0 < u_2 < u_3$ of $2u^3 - u^2 + b^{-2}$, the angle swept is

$$\Delta\phi(r_{\text{far}}) = \frac{4F(\psi|m)}{\sqrt{2(u_3 - u_1)}}, \quad m = \frac{u_2 - u_1}{u_3 - u_1}, \quad \sin^2 \psi = \frac{(u_3 - u_1)(u_2 - u_{\text{far}})}{(u_2 - u_1)(u_3 - u_{\text{far}})}, \quad (5)$$

which for $r_{\text{far}} \rightarrow \infty$ is Darwin's result [18, 19]. The reference uses the same r_{far} as the integration: comparing with the asymptotic angle would leave a systematic error of order b/r_{far}^2 . Equation (5), Darwin's form in terms of the periapsis and a direct 30-digit quadrature agree to double precision for $b \in [5.2, 10^4]$. TC1b takes $b = b_c(1 + 10^{-k})$, $k = 1, \dots, 10$, where the number of windings grows like $-\ln(b/b_c - 1)/2\pi$ (Bozza's strong-deflection limit [20]).

b. TC2, photon sphere. A photon starts at a turning point $r_0 = 3 + \delta_0$ with $b = b_c$ and is followed until $|r - 3| > 0.1$. Small offsets grow like $\cosh(\lambda_L t)$ with $\lambda_L = 1/(3\sqrt{3})$ [21], a factor

$e^{2\pi} \approx 535$ per orbit; the exact exit angle is an elliptic integral from the turning point. The equatorial orbit with $\delta_0 = 0$ is an exact fixed point of every Runge–Kutta map: the radial components of the vector field vanish and t, ϕ are linear in λ . It survived 900 orbits in a first test. The experiments therefore tilt the orbital plane by 45° , so that truncation errors in L^2 reach r .

c. TC3, circular orbits. Timelike circular orbits with $E = f/\sqrt{1 - 3/r_c}$, $L = \sqrt{r_c/(1 - 3/r_c)}$ and $\Omega = r_c^{-3/2}$. Like the photon sphere they are fixed points of the radial dynamics; their errors measure round-off, not truncation. TC3m starts on the ISCO ($r = 6$, $E = \sqrt{8/9}$, $L = 2\sqrt{3}$) in a plane inclined by 45° and stops when $|r - 6| > 1$.

d. TC4 and TC5, eccentric orbits. Bound orbits are labelled by the semi-latus rectum p and eccentricity e [22], with $r_p = p/(1 + e)$ and $r_a = p/(1 - e)$. The orbit equation $(du/d\phi)^2 = 2(u - u_1)(u - u_2)(u - u_3)$ has $u_1 = (1 - e)/p$, $u_2 = (1 + e)/p$ and $u_3 = \frac{1}{2} - 2/p$, which gives the angle per radial period and the orbit in closed form [23]:

$$\Phi = \frac{4K(m)}{\sqrt{2(u_3 - u_1)}}, \quad u(\phi) = u_1 + (u_2 - u_1) \operatorname{cd}^2\left(\sqrt{\frac{u_3 - u_1}{2}} \phi \middle| m\right), \quad m = \frac{u_2 - u_1}{u_3 - u_1}. \quad (6)$$

The radial periods in proper and coordinate time come from a 30-digit quadrature in the angle χ of $r = \frac{1}{2}(r_a + r_p) - \frac{1}{2}(r_a - r_p) \cos \chi$, which removes both turning-point singularities and gives Φ a second time (agreement 10^{-16}). The long-term case TC5 follows (20, 0.5) for 10^3 – 10^4 periods; its phase error is $\phi_N - N\Phi$ at the N -th periapsis, which needs no reference integration.

e. TC6, inclined orbits. The orbit (20, 0.5) in planes inclined by 30° , 60° and 85° . The references are L^2 and the orientation of the orbital plane, measured by the angle between the angular-momentum vector (L_x, L_y, L_z) at each sample and at the start.

f. Validation. Table II compares tightly integrated values (DOP853, tolerance 10^{-13} , formulation (b)) with every reference. All agree to 10^{-11} or better, except the cases amplified by the photon sphere, which agree to about 10^{-8} .

V. RESULTS

A. Convergence on geodesics

Every fixed-step method reaches its theoretical order on geodesics, in both formulations (Fig. 2, Table III). On the strong-field orbit (7.5, 0.5) the measured orders are within ± 0.05 of theory, on the deflection with $b = 6$ within ± 0.08 . GL1 and Tao2 are almost indistinguishable in (b), and GL1 is about 35 times less accurate in (a) than in (b) at the same step. The adaptive pairs show tolerance proportionality: the global error falls like $\text{tol}^{0.85-1.04}$ for DP5 and DOP853, on the same

TABLE II. Validation (T2): DOP853 at tolerance 10^{-13} in formulation (b) against the exact references. The last row compares with the value listed by the predecessor project, `schwarzschild-geodesics`: its “analytic” 2.127093985 was evaluated at the smallest sampled radius rather than the true periapsis. The exact value, from the closed form and a 40-digit quadrature, is 2.12709400813.

Check	Numerical	Analytical	Difference
TC1 deflection $\Delta\phi(r_{\text{far}})$, $b = 6$, $r_{\text{far}} = 1000$	4.84898	4.84898	6.1×10^{-13}
TC1 deflection $\Delta\phi(r_{\text{far}})$, $b = 40$, $r_{\text{far}} = 10000$	3.2417	3.2417	4.1×10^{-13}
TC1 deflection $\Delta\phi(r_{\text{far}})$, $b = 1000$, $r_{\text{far}} = 10000$	2.94527	2.94527	3.5×10^{-14}
TC1b windings, $b = b_c(1 + 1e-6)$	2.63346	2.63346	1.1×10^{-8}
TC2 orbits before $ r - 3 > 0.1$, $r_0 = 3 + 1e-6$	1.93916	1.93916	1.2×10^{-8}
TC2 growth rate of $ r - 3 $ in t (Lyapunov exponent)	0.192515	0.19245	6.5×10^{-5}
TC3 circular orbit $r_c = 10$: $d\phi/dt$	0.0316228	0.0316228	0
TC3 circular orbit $r_c = 6$: $d\phi/dt$	0.0680414	0.0680414	2.6×10^{-15}
TC4 periapsis advance, $(p, e) = (20, 0.5)$	1.23386	1.23386	5.6×10^{-13}
TC4 radial period in proper time, $(p, e) = (20, 0.5)$	930.547	930.547	8.0×10^{-11}
TC4 periapsis advance, $(p, e) = (7.5, 0.5)$	9.35519	9.35519	1.3×10^{-11}
TC4 radial period in proper time, $(p, e) = (7.5, 0.5)$	307.14	307.14	5.0×10^{-11}
TC6 inclined 60° : in-plane angle after 2 periods	15.0341	15.0341	1.6×10^{-11}
TC6 inclined 60° : tilt of the orbital plane (rad)	1.58374e-13	0	1.6×10^{-13}
TC6 inclined 60° : max relative change of L^2	1.35381e-13	0	1.4×10^{-13}
Old README: precession $r_{\text{apo}} = 20$, $L = 4.2$	2.12709	2.12709	2.3×10^{-8}

TABLE III. Measured order of convergence (T3), least-squares slope over the asymptotic range with its standard error.

Method	Theory	Deflection (a)	Deflection (b)	Orbit (a)	Orbit (b)
RK4	4	4.08 ± 0.02	3.96 ± 0.01	3.98 ± 0.01	4.01 ± 0.00
GL1	2	2.00 ± 0.00	2.00 ± 0.00	2.00 ± 0.00	2.00 ± 0.00
GL2	4	4.00 ± 0.00	3.98 ± 0.01	3.98 ± 0.01	4.00 ± 0.00
GL3	6	6.00 ± 0.00	6.06 ± 0.04	5.97 ± 0.01	6.05 ± 0.01
Tao2	2	–	2.00 ± 0.00	–	2.00 ± 0.00
Tao4	4	–	4.01 ± 0.00	–	4.00 ± 0.00

curves as SciPy’s (Fig. 3). On (20, 0.5) RK4 stays pre-asymptotic over the whole usable range (slopes 4.5 to 4.07 before round-off), which is why the orders are measured on the strong-field orbit.

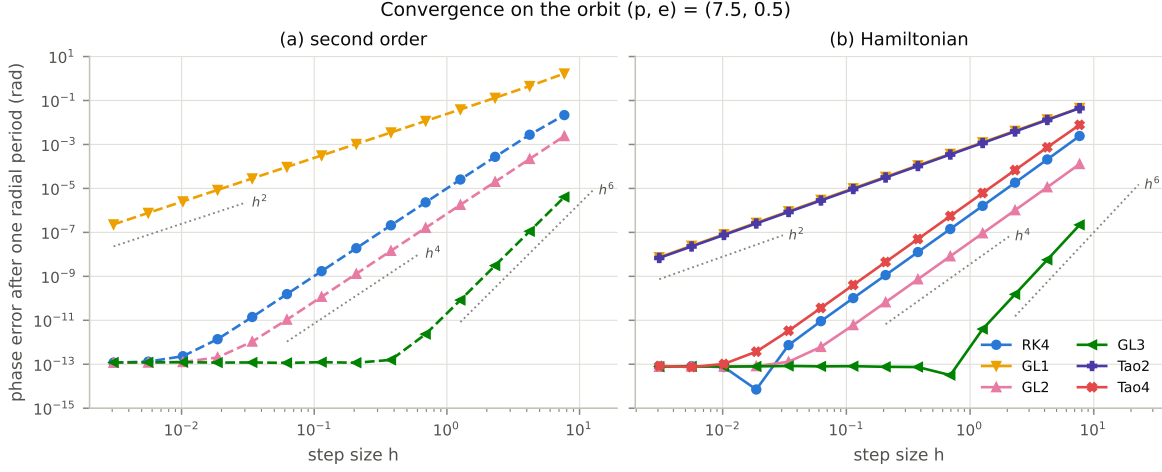


FIG. 2. Convergence of the fixed-step methods: error against step size for the deflection $b = 6$ (error in $\Delta\phi(r_{\text{far}})$) and one radial period of the orbit $(7.5, 0.5)$ (error in ϕ), in both formulations, with reference slopes 2, 4 and 6 (F4).

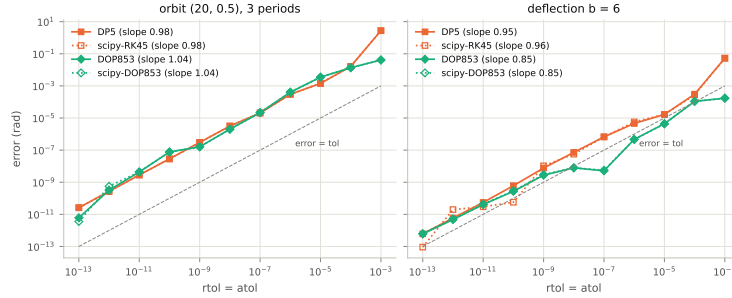


FIG. 3. Tolerance proportionality of the adaptive pairs: global error against $\text{rtol} = \text{atol}$, ours and SciPy's (F5).

B. Short integrations: cost per accuracy (RQ3)

Fig. 4 and Table IV compare the cost of reaching a given error on three short problems: the deflection $b = 6$, three periods of $(20, 0.5)$ and two periods of the same orbit inclined by 60° . **DOP853 is the cheapest method on every problem and at every accuracy.** To reach 10^{-10} in formulation (b) it needs $1.4\text{--}3.2 \times 10^3$ evaluations, against $1.6\text{--}3.6 \times 10^4$ for GL3, the best fixed-step method, and $3.5 \times 10^4\text{--}1.7 \times 10^5$ for RK4. Tao4 costs 1.2–10 times as much as RK4 for the same error, and GL1 and Tao2 never reach 10^{-10} in the sweep. Formulation (b) is the cheaper one for most pairs of method and problem, but not all: on the inclined orbit DOP853 and GL3 reach 10^{-10} more cheaply in (a).

The deflection angle itself is reproduced to 10^{-14} relative far from the hole and to 10^{-11} at

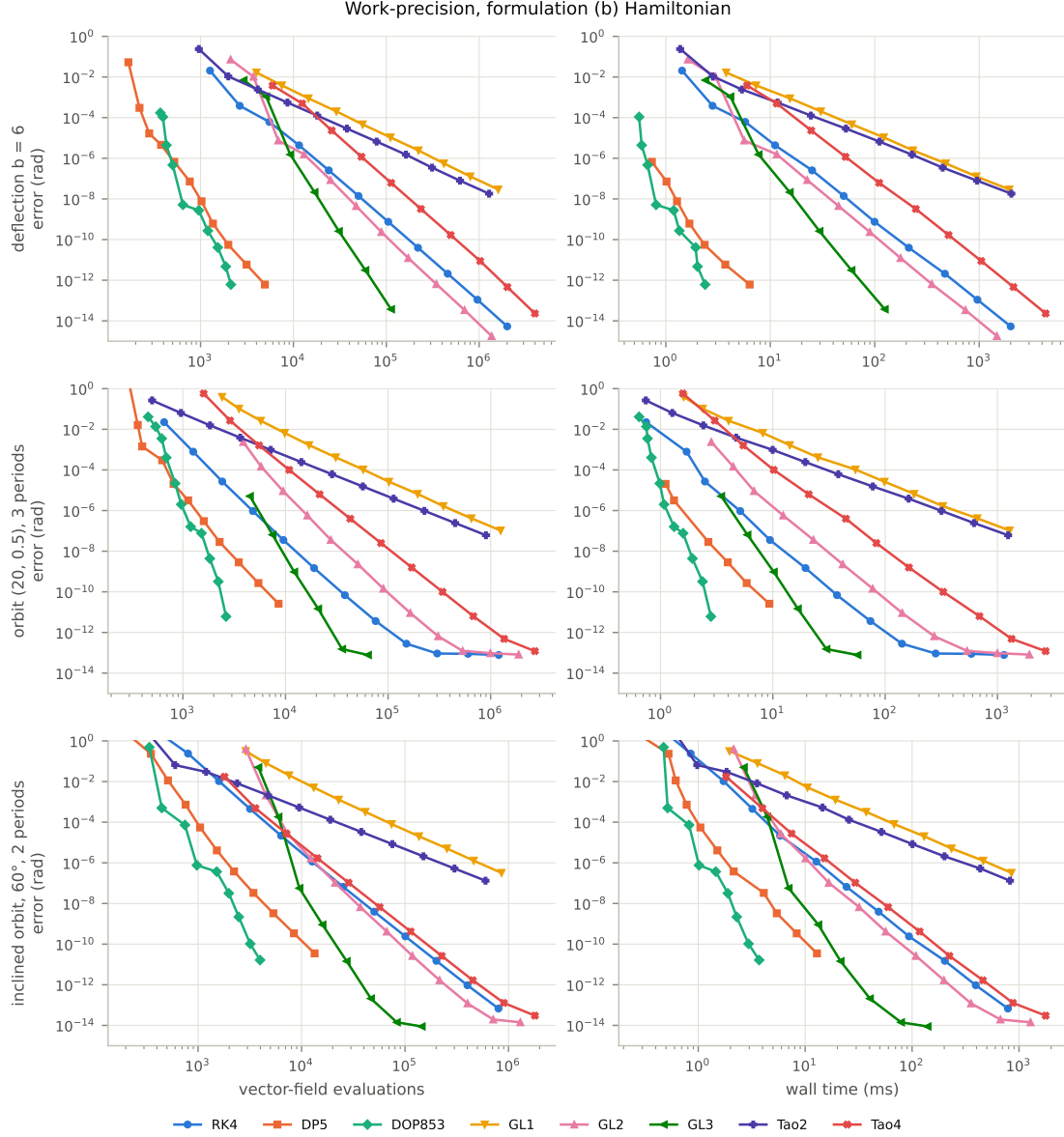


FIG. 4. Work-precision in formulation (b): error against vector-field evaluations (left) and wall time (right) for the three short problems (F6). Each curve is the Pareto front of a sweep over 12 step sizes or 11 tolerances.

$b - b_c = 10^{-3}$, where the problem amplifies errors (Fig. 5). The fourth-order weak-field series is good to 10^{-11} at $b \approx 2000$ and off by more than 1% below $b \approx 10$; Bozza's limit is accurate near b_c and fails above $b - b_c \approx 1$. Near the critical impact parameter the number of windings follows $-\ln(b/b_c - 1)/2\pi$, and the integration error grows exactly like the condition number $1/(b - b_c)$: $3.3 \times 10^{-13}/(b - b_c)$ windings in (b) and $1.4 \times 10^{-10}/(b - b_c)$ in (a) (Fig. 6). The periapsis advance agrees with $\Phi - 2\pi$ to 4×10^{-13} away from the separatrix and to 1.5×10^{-8} at $p = 7.02$, where the

TABLE IV. Evaluations needed to reach the error 10^{-6} or 10^{-10} (T5), interpolated along each method's work-precision front, in formulation (b) with (a) in parentheses. A dash means the target was not reached in the sweep.

Method	Deflection, 10^{-6}	10^{-10}	Orbit, 10^{-6}	10^{-10}	Inclined, 10^{-6}	10^{-10}
RK4	1.7×10^4 (1.7×10^4)	1.7×10^5 (1.6×10^5)	4.8×10^3 (8.3×10^3)	3.5×10^4 (8.4×10^4)	1.3×10^4 (1.2×10^4)	1.3×10^5 (1.3×10^5)
DP5	487 (1.3×10^3)	1.8×10^3 (7.1×10^3)	1.3×10^3 (2.5×10^3)	6.6×10^3 (–)	1.9×10^3 (1.6×10^3)	1.1×10^4 (–)
DOP853	476 (1.2×10^3)	1.4×10^3 (–)	1.0×10^3 (1.5×10^3)	2.3×10^3 (3.5×10^3)	958 (1.4×10^3)	3.2×10^3 (2.4×10^3)
GL1	3.3×10^5 (1.9×10^6)	–	4.3×10^5 (9.6×10^5)	–	5.2×10^5 (7.4×10^5)	–
GL2	1.4×10^4 (4.9×10^4)	1.1×10^5 (3.6×10^5)	1.5×10^4 (2.1×10^4)	9.7×10^4 (1.4×10^5)	1.4×10^4 (1.8×10^4)	8.9×10^4 (1.2×10^5)
GL3	9.8×10^3 (1.9×10^4)	3.6×10^4 (6.4×10^4)	5.4×10^3 (7.5×10^3)	1.6×10^4 (2.0×10^4)	8.1×10^3 (8.4×10^3)	2.1×10^4 (1.8×10^4)
Tao2	2.0×10^5 (–)	–	2.2×10^5 (–)	–	2.2×10^5 (–)	–
Tao4	5.6×10^4 (–)	5.6×10^5 (–)	3.4×10^4 (–)	3.4×10^5 (–)	1.6×10^4 (–)	1.6×10^5 (–)

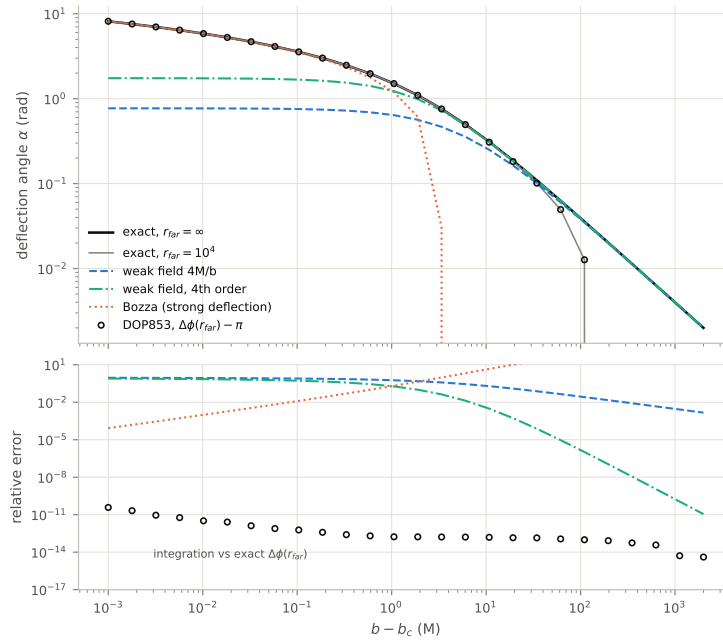


FIG. 5. Deflection angle against $b - b_c$: exact, weak-field series, Bozza's limit and the integration (DOP853, tolerance 10^{-13}); bottom, relative errors (F2).

orbit sweeps 3.4 extra turns per period and the weak-field value $6\pi/p$ is off by 87% (Fig. 6).

C. Long integrations: growth of the errors (RQ1)

Fig. 7 shows the mass-shell error over 10^4 radial periods of $(20, 0.5)$, with 1000 steps per period for the fixed-step methods and tolerance 10^{-10} for the adaptive pairs. The hypothesis of RQ1 holds without exception (Table V). **RK4, DP5 and DOP853 drift linearly:** the exponents

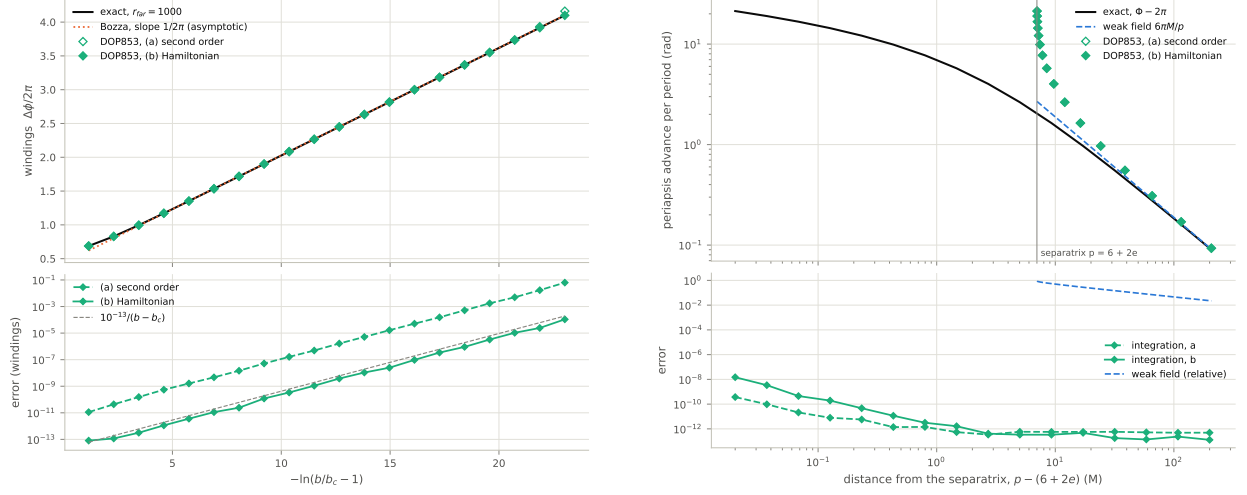


FIG. 6. Left: windings near the critical impact parameter and their integration error, which follows the condition number $1/(b - b_c)$ (F12). Right: periapsis advance against p , exact, integrated and weak-field (F13).

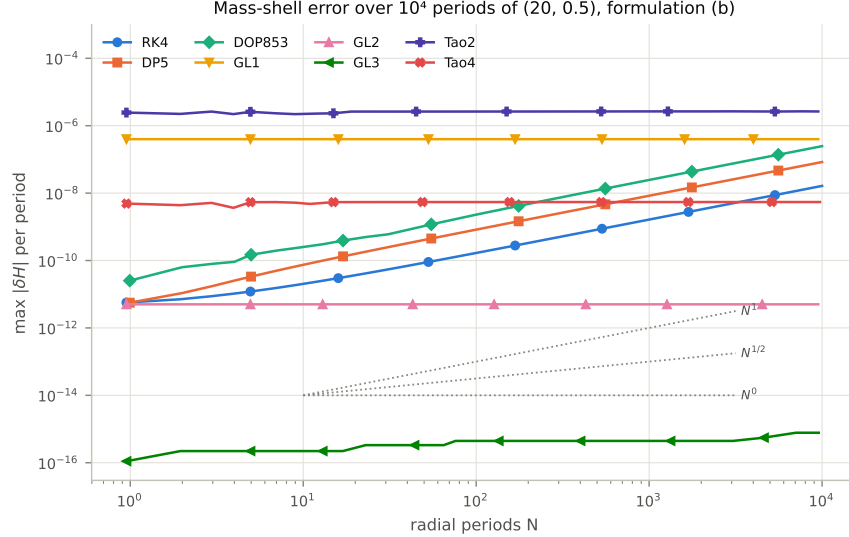


FIG. 7. Mass-shell error, maximum per radial period, over 10^4 periods of $(20, 0.5)$ in formulation (b), drawn as its maximum over logarithmic bins (F7).

fitted over the last decade are 1.000 ± 0.002 , and $|\delta H|$ reaches 1.6×10^{-8} (RK4) and 2.5×10^{-7} (DOP853). **GL1, GL2, Tao2 and Tao4 are bounded**, with exponents below 3×10^{-3} ; GL2 stays at 5.0×10^{-12} . GL3's truncation error is below round-off. Its error grows from 10^{-16} to 6.7×10^{-16} with exponent 0.48, Brouwer's random walk [24].

The phase error at the N -th periapsis (Fig. 8) grows quadratically for the Runge–Kutta methods

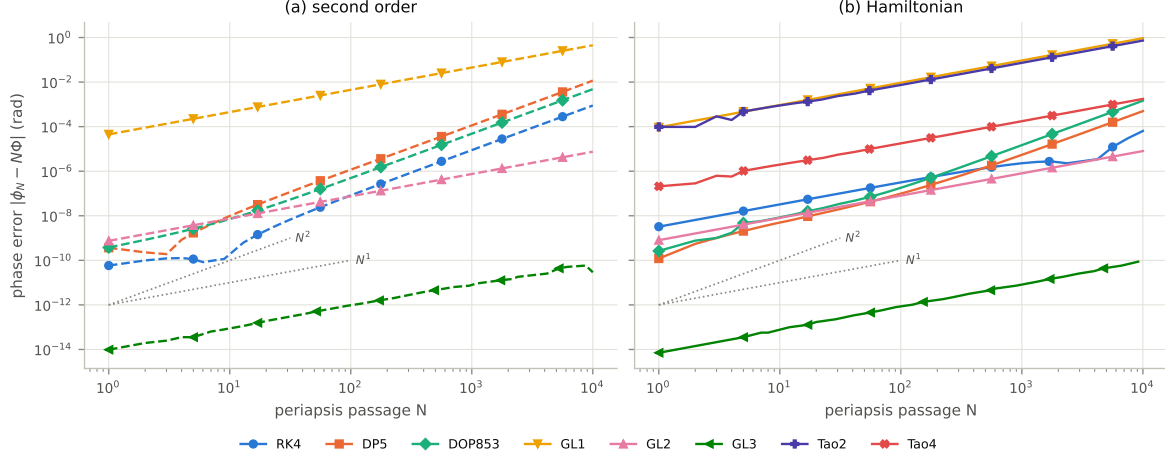


FIG. 8. Phase error $|\phi_N - N\Phi|$ at the N -th periapsis in both formulations, with slopes 1 and 2 (F9).

TABLE V. Growth laws over 10^4 periods of (20, 0.5) (T4): the mass-shell error in the last period, the log-log exponent over the last decade and its nearest law (0, 1/2, 1, 2), the same for the phase error, and the signed fit $\phi_N - N\Phi = c_1N + c_2N^2$ (radians per orbit and per orbit squared).

Method	$ \delta H $, last period	Exp. Law	Phase error	Exp. Law	c_1	c_2
RK4 (b)	1.6×10^{-8}	1.00 linear	6.4×10^{-5}	2.04 quadratic	3.2×10^{-9}	-9.6×10^{-13}
DP5 (b)	8.4×10^{-8}	1.00 linear	5.0×10^{-4}	1.97 quadratic	-4.9×10^{-10}	-4.9×10^{-12}
DOP853 (b)	2.5×10^{-7}	1.00 linear	1.5×10^{-3}	2.00 quadratic	2.4×10^{-10}	1.5×10^{-11}
GL1 (b)	4.0×10^{-7}	0.00 bounded	0.93	1.00 linear	9.3×10^{-5}	7.5×10^{-14}
GL2 (b)	5.0×10^{-12}	0.00 bounded	8.0×10^{-6}	1.00 linear	8.0×10^{-10}	-2.1×10^{-20}
GL3 (b)	6.7×10^{-16}	0.48 random walk	7.3×10^{-11}	1.02 linear	7.0×10^{-15}	2.4×10^{-20}
Tao2 (b)	2.6×10^{-6}	0.00 bounded	0.73	1.00 linear	-7.3×10^{-5}	1.2×10^{-12}
Tao4 (b)	5.2×10^{-9}	0.00 bounded	1.8×10^{-3}	1.00 linear	1.8×10^{-7}	-1.0×10^{-15}
RK4 (a)	8.7×10^{-10}	0.89 linear	8.9×10^{-4}	2.00 quadratic	6.7×10^{-11}	-8.9×10^{-12}
DP5 (a)	2.2×10^{-7}	1.00 linear	0.012	2.01 quadratic	-7.1×10^{-9}	1.2×10^{-10}
DOP853 (a)	9.4×10^{-8}	1.00 linear	4.8×10^{-3}	2.00 quadratic	-8.4×10^{-11}	-4.8×10^{-11}
GL1 (a)	4.0×10^{-7}	0.00 bounded	0.45	1.00 linear	-4.5×10^{-5}	8.8×10^{-14}
GL2 (a)	5.0×10^{-12}	0.00 bounded	7.5×10^{-6}	1.00 linear	-7.5×10^{-10}	2.2×10^{-18}
GL3 (a)	3.3×10^{-16}	0.25 random walk	2.9×10^{-11}	0.89 linear	-6.8×10^{-15}	2.3×10^{-19}

(exponents 1.97–2.04) and linearly for GL1, GL2 and Tao (1.000 ± 0.001). A log-log slope is a poor summary of a signed error, so Table V also fits $\phi_N - N\Phi = c_1N + c_2N^2$. For RK4 in (b) the constant frequency error c_1 and the drift c_2 have opposite signs and cancel near $N = 3364$, the dip in the figure. GL3's phase error, 7.3×10^{-11} at $N = 10^4$, is 5 ulp of $\phi \approx 7.3 \times 10^4$. Its linear growth is the spacing of floating-point numbers growing with ϕ .

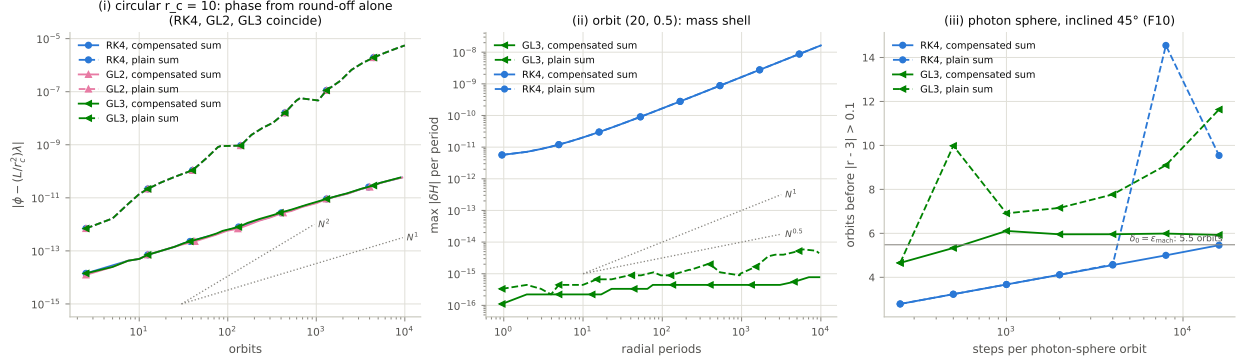


FIG. 9. Round-off with and without compensated summation (F19). (i) Phase of the circular orbit $r_c = 10$, where only round-off acts. (ii) Mass shell of GL3 and RK4 on $(20, 0.5)$. (iii) Survival on the photon sphere (Sec. VF).

a. Round-off. Compensated summation decides what happens once truncation errors are negligible (Fig. 9). On the circular orbit $r_c = 10$ every Runge–Kutta map is exact, so the error in ϕ after 10^4 orbits (10^7 steps) is round-off alone. With compensated summation it is 5.1×10^{-11} , 7 ulp of ϕ . With plain summation it is 5.5×10^{-6} and grows like $N^{2.05}$: the same increment is rounded the same way at every step, so the errors add coherently, and each is proportional to ϕ . GL3’s mass shell follows Brouwer’s law with compensation (exponent 0.48) and grows faster without it (0.67).

D. The role of the formulation (RQ2)

In formulation (b), E and L_z keep their initial values bit for bit in every sample of every method, Tao included, for all 10^4 periods. In (a) they are nonlinear functions of the state, and the Runge–Kutta methods drift linearly in E , L_z and the norm, while GL1 and GL2 stay bounded and GL3 stays at round-off (Fig. 10).

Gauss–Legendre in (a), where it is symmetric but not symplectic, conserves as well as in (b). The largest mass-shell error of GL2 is 5.0138×10^{-12} in both formulations, equal in every printed digit, and the phase drifts are -7.48×10^{-10} and $+8.03 \times 10^{-10}$ radians per orbit (Fig. 11). For this integrable, reversible system reversibility is enough, as the theory of reversible integrators predicts [6, Ch. XI]. RK4 drifts in both formulations.

The formulation matters most for rays that travel far. In (a) the angular momentum $L = r^2 u^\phi$ is rebuilt from a component that falls like r^{-2} , which an absolute tolerance cannot follow to full relative accuracy, so L and with it $b = L/E$ drift. At $b = 5.3$ and $r_{\text{far}} = 10^4$, DOP853 at tolerance

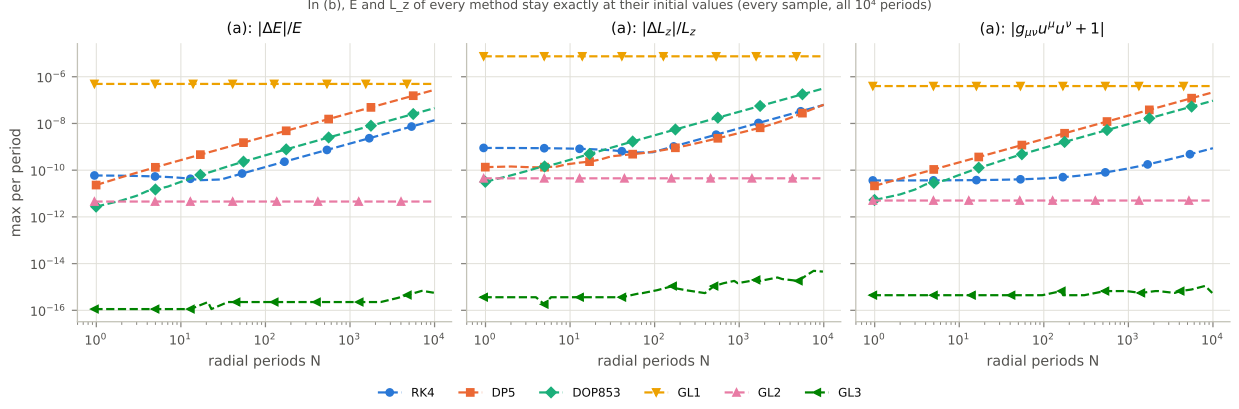


FIG. 10. Formulation (a): relative errors in E , L_z and the norm over 10^4 periods (F8). In (b), E and L_z do not change at all.

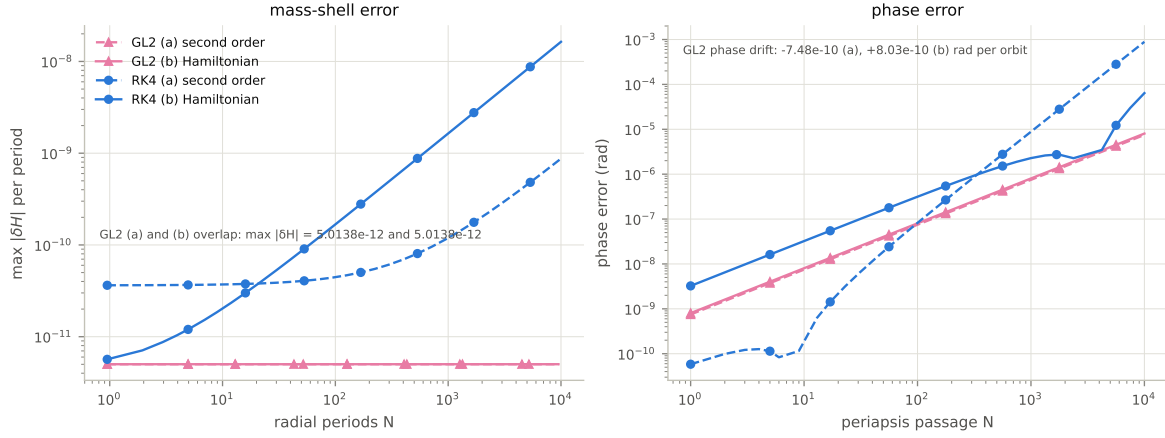


FIG. 11. GL2 in formulation (a), symmetric but not symplectic, against GL2 in (b), symplectic, and RK4 in both (F17).

10^{-13} makes an error of 1.2×10^{-8} in (a) and 3×10^{-12} in (b); near b_c the windings are 440 times less accurate in (a) (Fig. 6). At a fixed budget of 2×10^4 evaluations the adaptive pairs reach 10^{-11} – 10^{-12} in (b) for every b , but lose up to three orders near b_c in (a).

a. Inclined orbits. L^2 , the analogue of Carter’s constant, is bounded for GL2 and GL3 in both formulations and drifts linearly for RK4 and DOP853 (Fig. 12). The orbital plane turns, and the turn is almost entirely a spurious precession of the node: the inclination changes by at most 7×10^{-9} for the Gauss methods, as it must with L_z exact and L^2 bounded. The node precesses linearly in time for every method, symplectic or not. Exact Schwarzschild has no nodal precession at all, so a drift in energy cannot change the rate. Near the pole (85°) every method is under-resolved at 1000 steps per period, and (b) is the worse formulation: the node of GL2 drifts by

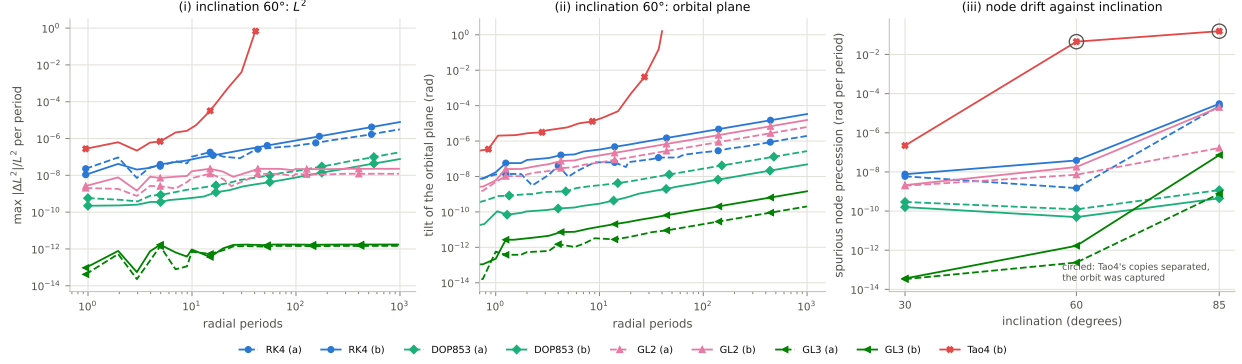


FIG. 12. Inclined orbits over 1000 periods (F18): (i) L^2 and (ii) tilt of the orbital plane at 60° ; (iii) spurious node precession against inclination. Tao4 fails at 60° and 85° (Sec. VG).

2.0×10^{-5} radians per period in (b) and 1.7×10^{-7} in (a).

E. Long integrations: cost per accuracy (RQ3)

Fig. 13 repeats the work-precision comparison over 1000 periods, where errors that grow with time have time to matter. To reach a phase error of 10^{-4} , DOP853 is still the cheapest method (706 evaluations per period in (b), against 2.1×10^3 for GL3; Table VI). **A phase error of 10^{-8} is reached by GL3 alone**, for 6.2×10^3 evaluations per period. DOP853 stops at 4.8×10^{-6} at tolerance 10^{-13} , the tightest a double-precision controller accepts, because its phase error grows like N^2 . For the mass shell, $|\delta H| \leq 10^{-10}$, DOP853 remains the cheapest. Over three periods (Table IV) the same orbit needed 776 evaluations per period from DOP853 to reach 10^{-10} in phase, and 5.4×10^3 from GL3. The ranking turns over with the length of the integration.

F. Unstable and marginally stable orbits (RQ4)

a. Photon sphere. A photon starts exactly at $r = 3$ in a plane inclined by 45° . Its only perturbation is the integration error δ_0 , amplified like $e^{\lambda L t}$ until $|r - 3| > 0.1$ after about $\ln(0.1/\delta_0)/2\pi$ orbits. If δ_0 is proportional to the tolerance, every decade of tolerance buys $\ln 10/2\pi = 0.37$ orbits; if $\delta_0 \propto h^p$, every decade of steps buys $0.37p$. Both predictions hold (Fig. 14, Table VII): the adaptive pairs gain 0.33–0.43 orbits per decade (DOP853 in (b): 0.33); RK4, GL2 and Tao4 gain 1.38–1.49 per decade of steps against 1.47 predicted; GL3 gains 2.26 against 2.20. Then GL3 saturates at 5.8–6.2 orbits whatever the step. That is the round-off limit, near the $\text{arccosh}(0.1/\epsilon_{\text{mach}})/2\pi = 5.5$ orbits of an offset of one machine epsilon. **No method keeps a photon on the photon sphere**

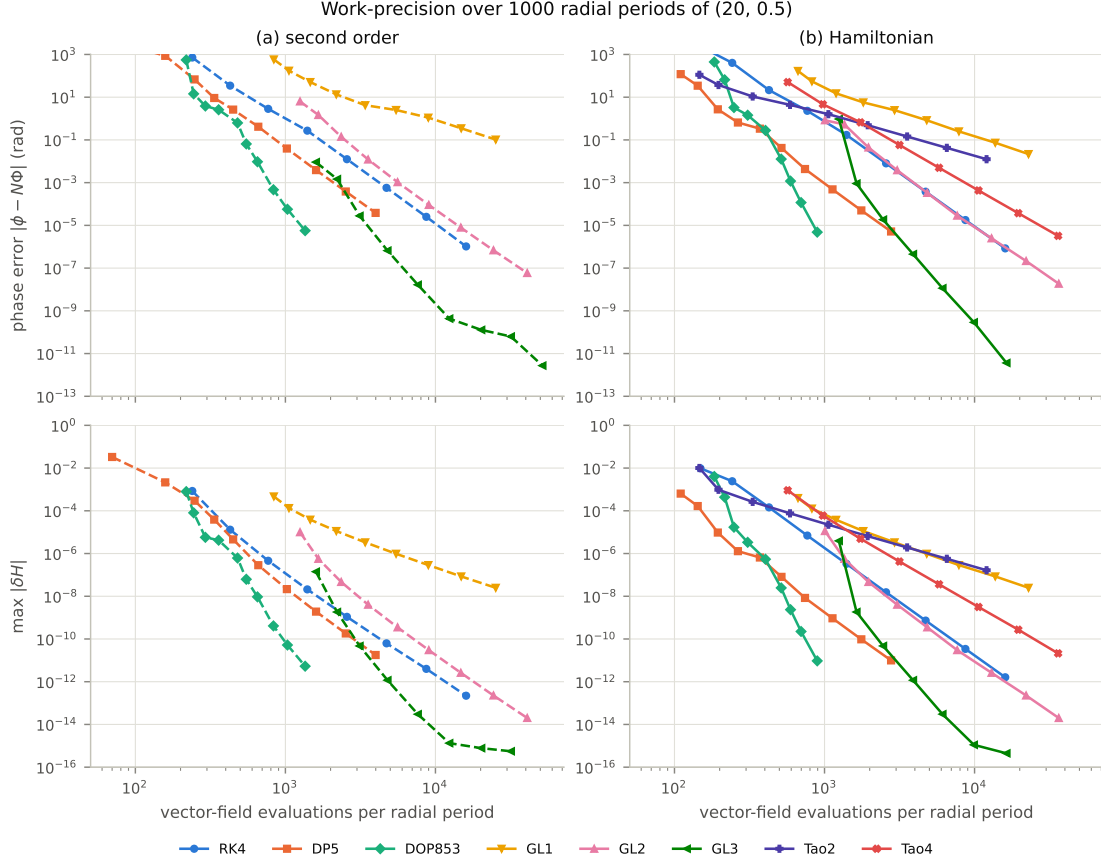


FIG. 13. Work-precision over 1000 radial periods of (20, 0.5): phase error and largest mass-shell error against evaluations per period (F20).

for more than about six orbits in double precision. Explicit offsets 10^{-2} – 10^{-15} follow the elliptic-integral prediction to 0.01 orbits down to 10^{-14} .

Plain floating-point summation makes the photon sphere look more stable than it is (Fig. 9, panel iii). With RK4 at 8000 steps per orbit the orbit survives 14.5 orbits with plain summation, against 5.0 with compensated summation. An increment of r smaller than half an ulp of 3 is lost, r stays at exactly 3, and the radial force, which is proportional to $r - 3$, stays exactly zero: rounding freezes the instability, the longer the smaller the step. A longer survival is not always a better integrator.

b. ISCO. At the ISCO the effective potential has an inflection point. With $r = 6 + \delta$ and $L^2 = 12$, $V_{\text{eff}} = 8/9 + \delta^3/1944 + \mathcal{O}(\delta^4)$, and an integration error acts as a small constant force ε :

$$\ddot{\delta} = \varepsilon - \frac{\delta^2}{1296}. \quad (7)$$

TABLE VI. Evaluations per radial period needed over 1000 periods of (20, 0.5) (T5b), formulation (b) with (a) in parentheses.

Method	Phase 10^{-4}	Phase 10^{-8}	$ \delta H 10^{-10}$
RK4	6.2×10^3 (6.6×10^3)	$- 7.0 \times 10^3$ (4.3×10^3)	
DP5	1.5×10^3 (3.3×10^3)	$- 1.7 \times 10^3$ (2.8×10^3)	
DOP853	706 (974)	–	743 (964)
GL1	–	–	–
GL2	6.1×10^3 (8.9×10^3)	$- 6.1 \times 10^3$ (7.1×10^3)	
GL3	2.1×10^3 (2.8×10^3)	6.2×10^3 (8.2×10^3)	2.3×10^3 (2.9×10^3)
Tao2	–	–	–
Tao4	1.5×10^4	–	2.5×10^4

TABLE VII. Photon-sphere survival (T6): orbits gained per decade of tolerance or of steps per orbit, against the prediction, and the orbits survived at the finest setting.

Method	Varied	Orbits per decade	Predicted	Orbits, finest
DOP853 (a)	tolerance	0.35	0.37	4.66
DOP853 (b)	tolerance	0.33	0.37	4.59
DP5 (a)	tolerance	0.43	0.37	4.82
DP5 (b)	tolerance	0.37	0.37	4.94
GL2 (a)	step size	1.41	1.47	4.91
GL2 (b)	step size	1.44	1.47	4.75
GL3 (a)	step size	2.49	2.20	5.78
GL3 (b)	step size	2.26	2.20	5.96
RK4 (a)	step size	1.49	1.47	4.65
RK4 (b)	step size	1.45	1.47	4.56
Tao4 (b)	step size	1.38	1.47	3.84

For $\varepsilon > 0$ a stable equilibrium appears at $\delta = 36\sqrt{\varepsilon}$, and the orbit oscillates about it without plunging. For $\varepsilon < 0$ it drifts inward under a constant acceleration until $\delta \sim -36\sqrt{|\varepsilon|}$ and then runs away in a finite time. Both stages take a time of order $|\varepsilon|^{-1/4}$. The growth is algebraic, and the sign of the error decides whether the orbit plunges at all. With $\varepsilon \propto h^p$ the time to plunge should scale as $n^{p/4}$ in the steps per orbit n , and as $\text{tol}^{-1/4}$ for the adaptive pairs. The measurements agree (Fig. 15): the slope is 1.11 for RK4 in (b), 1.09 for GL2 and 1.53 for GL3 (predicted 1, 1 and 1.5), and -0.252 against tolerance for DP5 in (b). The runs that survive oscillate on the

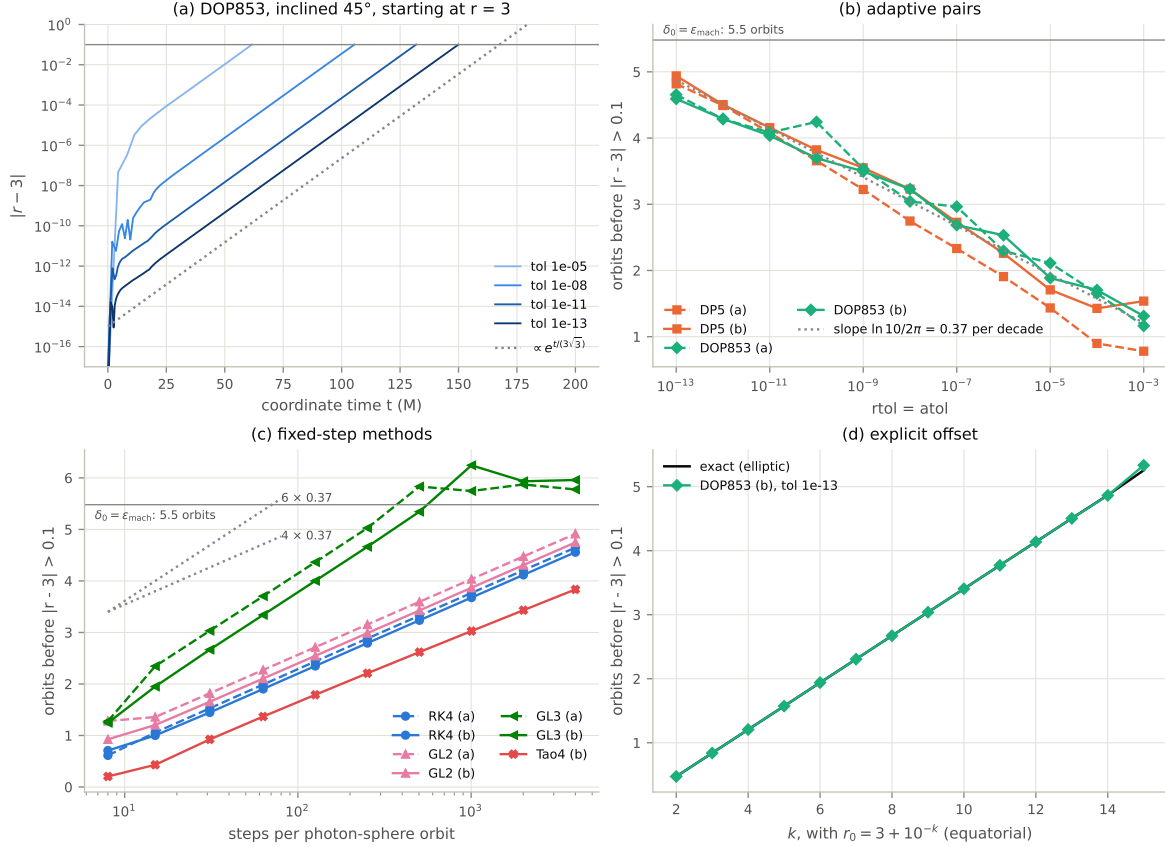


FIG. 14. Photon sphere (F10). (a) $|r - 3|$ against coordinate time for DOP853 at four tolerances, with the growth $e^{t/3\sqrt{3}}$. (b), (c) Orbits survived against tolerance and against steps per orbit. (d) Equatorial orbits from an explicit offset against the exact prediction.

outer side, as predicted: RK4 in (a) at 100 steps per orbit stays within $6.000 < r < 6.073$ for 2000 orbits. DOP853 never plunges at tolerances up to 10^{-6} ; DP5 in (b) plunges at 10^{-9} – 10^{-13} but not at 10^{-6} – 10^{-8} .

G. Method-specific findings

a. Tao's coupling ω . The error of Tao4 grows in proportion to ω once $\omega h \gtrsim 10^{-3}$, and the method diverges for $\omega h \gtrsim 1.4$; too small an ω lets the copies separate, by up to 23 within 300 periods for $\omega = 10^{-5}$ (Fig. 16). We use $\omega = 10^{-3}$ throughout: within a factor 1.2–5 of the best value on the orbits tested, with the copies within 3×10^{-6} over 300 periods. This choice fails on two of the long experiments. On the ISCO the copies separate by 1.7 in r and the orbit plunges within six orbits whatever the step; $\omega = 0.1$ holds them together. On the inclined orbits the run is captured after 41 periods at 60° and after 3 at 85° ; a larger ω helps at 60° but leaves L^2 drifting

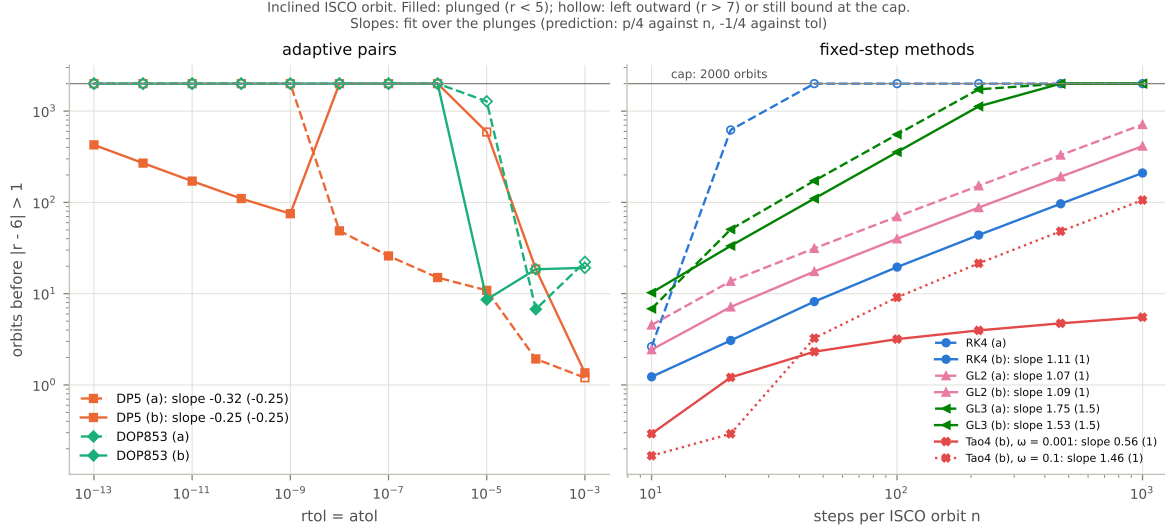


FIG. 15. Inclined ISCO orbit: orbits until $|r-6| > 1$ against tolerance and against steps per orbit, with the slopes fitted over the plunges (F11). Filled markers plunged, hollow ones left outward or were still bound after 2000 orbits.

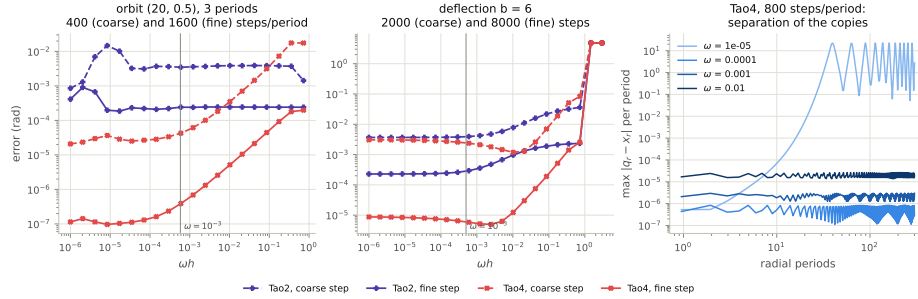


FIG. 16. Tao4: error against the coupling ω at fixed step, and the separation of the two copies (F14).

3000 times more than with GL2, and no ω between 10^{-3} and 1 works at 85° . The coupling is a problem-dependent parameter, and choosing it is a real cost of the method.

b. Implicit iteration. Iterations per step grow from 2–5 at small steps to 10–23 at the largest; GL3 needs the fewest, and (a) needs more than (b) (Fig. 17). Stopping the iteration at a tolerance instead of at stagnation makes GL2 drift: over 1000 periods $|\delta H|$ grows tenfold with tolerances 10^{-6} and 10^{-8} , threefold with 10^{-10} , and not at all with 10^{-12} or at stagnation. A tolerance of 10^{-12} needs 3.5 iterations per step against 6.7 for stagnation: the round-off rule is safe but not the cheapest safe rule.

c. Adaptive steps. On the deflection the step grows roughly in proportion to r , and DOP853 needs three to four times fewer steps than DP5 at the same tolerance (Fig. 18).

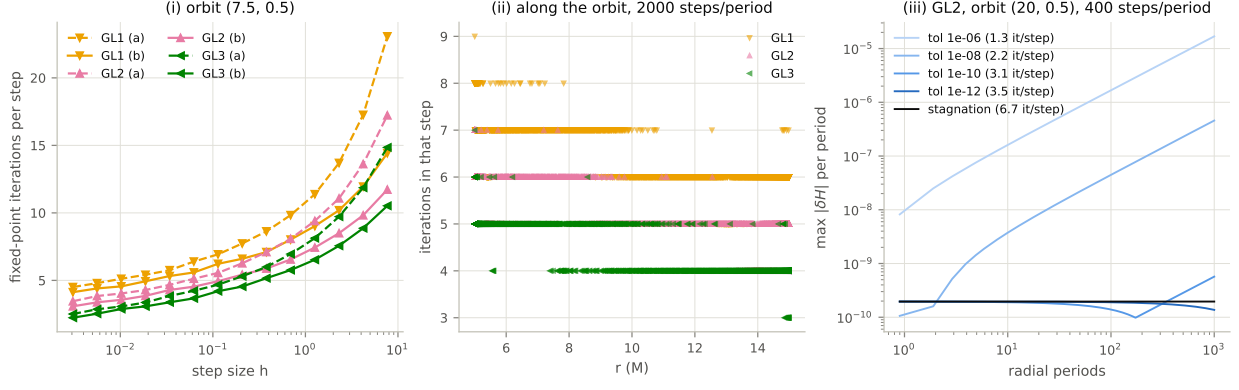


FIG. 17. Gauss–Legendre iteration (F15): iterations per step against h and along the orbit, and the drift of $|\delta H|$ when the iteration stops at a tolerance.

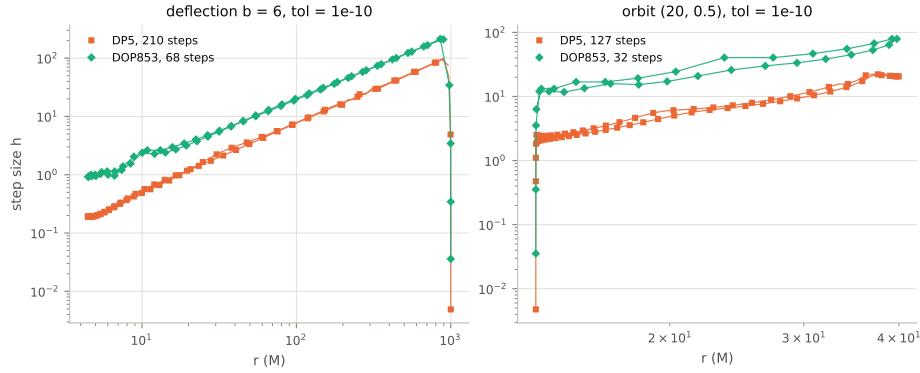


FIG. 18. Step sizes of the adaptive pairs against r on the deflection and on an eccentric orbit (F16).

VI. DISCUSSION AND RECOMMENDATIONS

a. Answers to the research questions. RQ1: the Runge–Kutta methods, fixed-step or adaptive, drift linearly in the mass shell and quadratically in phase. The symplectic methods keep the mass-shell error bounded at $\mathcal{O}(h^p)$ and the phase error linear, and once their truncation error falls below round-off, compensated summation gives Brouwer’s $N^{1/2}$. RQ2: in Hamilton’s form E and L_z are exact for every method; in the second-order form they drift for every non-geometric method. Formulation (b) is up to 440 times more accurate for rays that go far out. Symmetry alone gives Gauss–Legendre in (a) the same bounded errors as in (b). The one case that favours (a) is the near-polar orbit, where the spurious node precession is about 100 times smaller. RQ3: DOP853 is the cheapest method over a few periods at every accuracy and over 1000 periods for modest accuracy; for a phase error of 10^{-8} over 1000 periods only GL3 delivers. Tao4 is never competitive. RQ4: on the photon sphere accuracy buys survival logarithmically, $\ln 10/2\pi$ orbits

TABLE VIII. Recommendations (T7), with the evidence they rest on.

Task	Recommendation	Evidence
Ray tracing, short integrations (deflection, a few orbits)	DOP853 in formulation (b)	Cheapest at every accuracy (Table IV); exact E , L_z and up to 440 times more accurate than (a) near b_c (Fig. 6).
Long orbit evolution where the phase matters (10^3 periods and more)	GL3 in (b), iterated to stagnation (or to 10^{-12}), with compensated summation	The only method to reach a phase error of 10^{-8} over 1000 periods (Table VI); bounded invariants (Fig. 7); Brouwer's law (Fig. 9).
Long runs at modest accuracy	DOP853 is cheaper, but its invariants drift linearly	Table VI, Fig. 7, Fig. 8.
Studies that need bounded invariants (Poincaré sections, statistics of chaotic orbits)	A Gauss method with fixed step, in either formulation	Bounded H , L^2 in (a) and (b) (Fig. 11, Fig. 12).
Unstable orbits (photon sphere)	Any accurate method with compensated summation; expect at most ~ 6 orbits	Fig. 14, Table VII; plain summation fakes survival (Fig. 9).
Explicit symplectic integration of a general metric	Tao's method only with ω tuned for the problem at hand	Fails on the ISCO and on inclined orbits with ω tuned on a stable orbit (Sec. V G).

per decade, up to about six orbits in double precision for every method. At the ISCO the time to plunge grows algebraically, as $|\varepsilon|^{-1/4}$, and an error of the stabilizing sign prevents the plunge.

b. Recommendations. Table VIII turns these answers into advice (RQ5). Two conclusions go beyond the hypotheses we started from. First, the cost advantage of the adaptive pair is large enough that geometric integration pays off only when the integration is long *and* the phase matters. Over 10^3 periods a mass-shell error of 10^{-10} is cheaper to buy with a tight tolerance than with a symplectic method, although the drift of DOP853 will overtake the bounded error of GL3 on longer runs. Second, the round-off details matter: compensated summation turns a quadratic phase drift into a few ulp, and its absence can make an integration look more stable than the physics allows.

c. Limitations. The study uses fixed steps for the geometric methods; variable steps destroy their long-time properties unless the time is transformed [6, 25], and Ref. [7] shows one way to combine them. Only Schwarzschild is covered, where t and ϕ are cyclic and the problem is integrable; in nonintegrable systems (charged or spinning particles, perturbed metrics) the advantage of symplectic methods should be larger. Wall times compare our own implementations, whose generic step code costs 2–5 times the bare vector field, so they are not a statement about the algorithms’ intrinsic cost. The explicit symplectic splittings of Wang et al. [8] and Taylor methods [26] were not tested.

d. Extension to Kerr. The metric interface takes the full inverse metric and its derivatives, and none of the integrators or formulations knows the metric. Adding Kerr in Boyer–Lindquist coordinates is a new metric class, analytic references (Carter’s constant, spherical photon orbits) and new test cases, with no change to the integrator code. The inclined orbits of TC6 already exercise the code path that will monitor Carter’s constant.

VII. CONCLUSIONS

We compared eight integrators on Schwarzschild geodesics in two formulations, against exact references, over integrations from a single deflection to 10^4 orbits. The classical expectations of geometric integration hold quantitatively: linear against quadratic phase error, linear drift against bounded invariants, Brouwer’s law at round-off, and symmetric methods behaving like symplectic ones on this reversible, integrable system. The practical picture is less one-sided than the theory suggests. A high-order adaptive Runge–Kutta pair is the right tool for ray tracing and for long runs at modest accuracy. Sixth-order Gauss–Legendre is the right tool when the phase must stay accurate over thousands of orbits. Tao’s explicit symplectic method needs a coupling tuned to each problem.

Two results concern orbits at the edge of stability. On the photon sphere, accuracy buys survival only logarithmically, and double precision caps it at about six orbits for every method. At the ISCO the integration error decides, through its sign, whether an orbit plunges at all, and through its size, as $|\varepsilon|^{-1/4}$, when. In both cases the rounding of the state update, compensated or not, changes the answer qualitatively.

All code, cached summaries, figures and this report are produced by `make all` from a clean checkout. The interface between metric and integrator is ready for the Kerr metric, the next step of this work, and for its (anti-)de Sitter generalizations.

ACKNOWLEDGMENTS

The numerical work uses NumPy [27], SciPy [28], Numba [29], mpmath [30] and Matplotlib [31]. Code, data and the scripts that produce every figure and table are available at <https://github.com/NugiePhysics/geodesic-integrators>.

Appendix A: Christoffel symbols and Hamilton's equations

The nine independent non-zero Christoffel symbols of the Schwarzschild metric are

$$\begin{aligned}\Gamma_{tr}^t &= \frac{1}{r(r-2)}, & \Gamma_{tt}^r &= \frac{r-2}{r^3}, & \Gamma_{rr}^r &= -\frac{1}{r(r-2)}, & \Gamma_{\theta\theta}^r &= -(r-2), \\ \Gamma_{\phi\phi}^r &= -(r-2)\sin^2\theta, & \Gamma_{r\theta}^\theta &= \Gamma_{r\phi}^\phi = \frac{1}{r}, & \Gamma_{\phi\phi}^\theta &= -\sin\theta\cos\theta, & \Gamma_{\theta\phi}^\phi &= \cot\theta.\end{aligned}$$

The code writes them in terms of $r-2$ rather than $f = 1 - 2/r$, which keeps every component accurate to full relative precision down to the horizon. Hamilton's equations for (2) are

$$\begin{aligned}\dot{t} &= -\frac{p_t}{f}, & \dot{r} &= fp_r, & \dot{\theta} &= \frac{p_\theta}{r^2}, & \dot{\phi} &= \frac{p_\phi}{r^2\sin^2\theta}, & \dot{p}_t &= \dot{p}_\phi = 0, \\ \dot{p}_r &= -\frac{f'}{2} \left(\frac{p_t^2}{f^2} + p_r^2 \right) + \frac{1}{r^3} \left(p_\theta^2 + \frac{p_\phi^2}{\sin^2\theta} \right), & \dot{p}_\theta &= \frac{\cos\theta}{r^2\sin^3\theta} p_\phi^2.\end{aligned}$$

Both sets were checked against a symbolic derivation from $g_{\mu\nu}$ evaluated at 50 digits, to 6×10^{-16} relative per component for $r-2$ from 10^{-8} to 10^4 .

Appendix B: Gauss–Legendre methods

The s -stage Gauss method has nodes c_i at the zeros of the shifted Legendre polynomial of degree s , and $a_{ij} = \int_0^{c_i} \ell_j(\tau) d\tau$, $b_j = \int_0^1 \ell_j(\tau) d\tau$ with the Lagrange polynomials ℓ_j on the nodes. For $s = 2$,

$$\begin{array}{c|cc} \frac{1}{2} - \frac{\sqrt{3}}{6} & \frac{1}{4} & \frac{1}{4} - \frac{\sqrt{3}}{6} \\ \frac{1}{2} + \frac{\sqrt{3}}{6} & \frac{1}{4} + \frac{\sqrt{3}}{6} & \frac{1}{4} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array} \quad \text{and for } s = 3, \quad \begin{array}{c|ccc} \frac{1}{2} - \frac{\sqrt{15}}{10} & \frac{5}{36} & \frac{2}{9} - \frac{\sqrt{15}}{15} & \frac{5}{36} - \frac{\sqrt{15}}{30} \\ \frac{1}{2} & \frac{5}{36} + \frac{\sqrt{15}}{24} & \frac{2}{9} & \frac{5}{36} - \frac{\sqrt{15}}{24} \\ \frac{1}{2} + \frac{\sqrt{15}}{10} & \frac{5}{36} + \frac{\sqrt{15}}{30} & \frac{2}{9} + \frac{\sqrt{15}}{15} & \frac{5}{36} \\ \hline & \frac{5}{18} & \frac{4}{9} & \frac{5}{18} \end{array}$$

GL1 is the implicit midpoint rule ($c = a = \frac{1}{2}$, $b = 1$). One step solves $Z_i = h \sum_j a_{ij} F(y_n + Z_j)$ by fixed-point iteration and sets $y_{n+1} = y_n + \sum_i d_i Z_i$ with $d^\top = b^\top A^{-1}$, which avoids a final evaluation of F . The start value of Z is the collocation polynomial of the previous step, evaluated at the new

nodes. The iteration stops when $\|\Delta Z\|$ no longer decreases and is within 100 ulp of $\|Z\|$ (Sec. III), or after 60 iterations, which never occurred in the experiments.

Appendix C: Reproducibility and computing environment

All integrators, metrics and formulations are compiled with Numba [29]. One compiled driver per kind of method receives the vector field as a typed first-class function, so that compilation happens once per machine (about 60 s) and is cached. Every run is described by a specification (test case, formulation, method, options) whose hash keys a cache of result rows that carry the git commit, library versions and CPU. Long runs are reduced to per-period maxima before caching. The command `make all` regenerates every figure and table from an empty cache and then compiles this report; on the machine of Table IX it takes about 23 minutes on four cores. Repeating it reproduced every summary table and figure byte for byte, except the wall times, which agreed to within 25% (median ratio 0.98).

TABLE IX. Computing environment of the results (T8).

Item	Value
CPU	Intel(R) Core(TM) i5-7300HQ CPU @ 2.50GHz
logical cores	4
operating system	Linux-7.0.0-34-generic-x86_64-with-glibc2.43
Python	3.14.7
numpy	2.5.3
scipy	1.18.1
numba	0.68.0
llvmlite	0.50.0
mpmath	1.3.0
matplotlib	3.11.2
pandas	3.0.6
geoint	1.0.0
threads per process	1 (NUMBA_NUM_THREADS = OMP_NUM_THREADS = 1)
parallel sweeps	up to 4 forked worker processes; timing runs one at a time
timing	median of 2 repeats after a warm-up run (ADR 0003)
floating point	IEEE 754 double; compensated summation in fixed-step updates

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